Midterm Review

ECE 447: Data Analysis and Machine Learning for Engineers

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Simplest Measure to Evaluate Your Model

Misclassification Rate

$$\text{misclassification rate} = \frac{\text{number of incorrect predictions}}{\text{total predictions}}$$

Confusion Matrix

		predictions	
		positive	negative
target	positive	**TP**	**FN**
	negative	**FP**	**TN**

Confusion Matrix Explained (I)

		predictions	
		positive	negative
target	positive	TP	FN
	negative	FP	TN

True Positive (TP)

Data point from test set, with positive target, is predicated to have a positive target.

True Negative (TN)

Data point from test set, with negative target, is predicated to have a negative target.

Confusion Matrix Explained (II)

		predictions	
		positive	negative
target	positive	**TP**	**FN**
	negative	**FP**	**TN**

False Positive (FP)

Data point from test set, with negative target, is predicated to have a positive target.

False Negative (FN)

Data point from test set, with positive target, is predicated to have a negative target.

Misclassification Rate in Terms of TP, TN, FP, and FN

Misclassification Rate

$$\text{misclassification rate} = \frac{\text{FP} + \text{FN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

Classification Accuracy

$$\text{classification accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

More Performance Measures for Categorical Targets (I)

Highlight different aspects of the performance.

True Positive Rate

Out of the positive targets, how many are classified correctly.

$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

		predictions	
		positive	negative
target	positive	TP	FN
	negative	FP	TN

True Negative Rate (a.k.a Specificity)

Out of the negative targets, how many are classified correctly.

$$\text{TNR} = \frac{\text{TN}}{\text{TN} + \text{FP}}$$

		predictions	
		positive	negative
target	positive	TP	FN
	negative	FP	TN

More Performance Measures for Categorical Targets (II)

False Positive Rate

Out of the negative targets, how many are classified incorrectly.

$$\text{FPR} = \frac{\text{FP}}{\text{TN} + \text{FP}}$$

		predictions	
		positive	negative
target	positive	**TP**	**FN**
	negative	**FP**	**TN**

False Negative Rate

Out of the positive targets, how many are classified incorrectly.

$$\text{FNR} = \frac{\text{FN}}{\text{TP} + \text{FN}}$$

		predictions	
		positive	negative
target	positive	**TP**	**FN**
	negative	**FP**	**TN**

Precision and Recall

Precision

Captures how often, when a model makes a positive prediction, the prediction turns out to be correct.

$$\frac{\text{TP}}{\text{TP} + \text{FP}}$$

		predictions	
		positive	negative
target	positive	TP	FN
	negative	FP	TN

Recall (a.k.a TPR, a.k.a Sensitivity)

How confident we can be that all the instances with the positive target level have been found by the model.

$$\frac{\text{TP}}{\text{TP} + \text{FN}}$$

		predictions	
		positive	negative
target	positive	TP	FN
	negative	FP	TN

In-Class Example

Given the following sample test set, compute each of the metrics introduced in previous slides.

ID	Target	Pred.	Outcome
1	spam	ham	FN
2	spam	ham	FN
3	ham	ham	TN
4	spam	spam	TP
5	ham	ham	TN
6	spam	spam	TP
7	ham	ham	TN
8	spam	spam	TP
9	spam	spam	TP
10	spam	spam	TP

ID	Target	Pred.	Outcome
11	ham	ham	TN
12	spam	ham	FN
13	ham	ham	TN
14	ham	ham	TN
15	ham	ham	TN
16	ham	ham	TN
17	ham	spam	FP
18	spam	spam	TP
19	ham	ham	TN
20	ham	spam	FP

Misclassification Rate

ID	Target	Pred.	Outcome
1	spam	ham	FN
2	spam	ham	FN
3	ham	ham	TN
4	spam	spam	TP
5	ham	ham	TN
6	spam	spam	TP
7	ham	ham	TN
8	spam	spam	TP
9	spam	spam	TP
10	spam	spam	TP

ID	Target	Pred.	Outcome
11	ham	ham	TN
12	spam	ham	FN
13	ham	ham	TN
14	ham	ham	TN
15	ham	ham	TN
16	ham	ham	TN
17	ham	spam	FP
18	spam	spam	TP
19	ham	ham	TN
20	ham	spam	FP

We have 20 examples, and out of which, 5 are incorrect. Hence, misclassification rate is $\frac{5}{20} = 0.25$.

Classification rate is 0.75.

Confusion Matrix

ID	Target	Pred.	Outcome
1	spam	ham	FN
2	spam	ham	FN
3	ham	ham	TN
4	spam	spam	TP
5	ham	ham	TN
6	spam	spam	TP
7	ham	ham	TN
8	spam	spam	TP
9	spam	spam	TP
10	spam	spam	TP

ID	Target	Pred.	Outcome
11	ham	ham	TN
12	spam	ham	FN
13	ham	ham	TN
14	ham	ham	TN
15	ham	ham	TN
16	ham	ham	TN
17	ham	spam	FP
18	spam	spam	TP
19	ham	ham	TN
20	ham	spam	FP

		predictions	
		spam	ham
target	spam	6	3
	ham	2	9

Lets take a look here to know what kind of mistakes the model is making

Precision and Recall

		predictions	
		spam	ham
target	spam	6	3
	ham	2	9

Precision: How often emails marked as spam actually are spam.

$$\text{precision} = \frac{6}{(6+2)} = 0.750$$

Recall: How often the spam messages in the set were actually marked as *spam.*

$$\text{recall} = \frac{6}{(6+3)} = 0.667$$

More Interpretation

		predictions	
		spam	ham
target	spam	6	3
	ham	2	9

How likely it is that a genuine *ham* email could be marked as *spam*

$$1 - \text{precision} = 25\,\%$$

How likely it is that a spam email will be missed by the system and end up in our inbox.

$$1 - \text{recall} = 33.333\,\%$$

Precision and Recall for Every Class in Multinomial Targets

Precision for class l

$$\text{precision}(l) = \frac{\text{TP}(l)}{\text{TP}(l) + \text{FP}(l)}$$

Recall for class l

$$\text{recall}(l) = \frac{\text{TP}(l)}{\text{TP}(l) + \text{FN}(l)}$$

Prediction Scores

Many classification models provide not *target value* but a *prediction score*

0 or 1

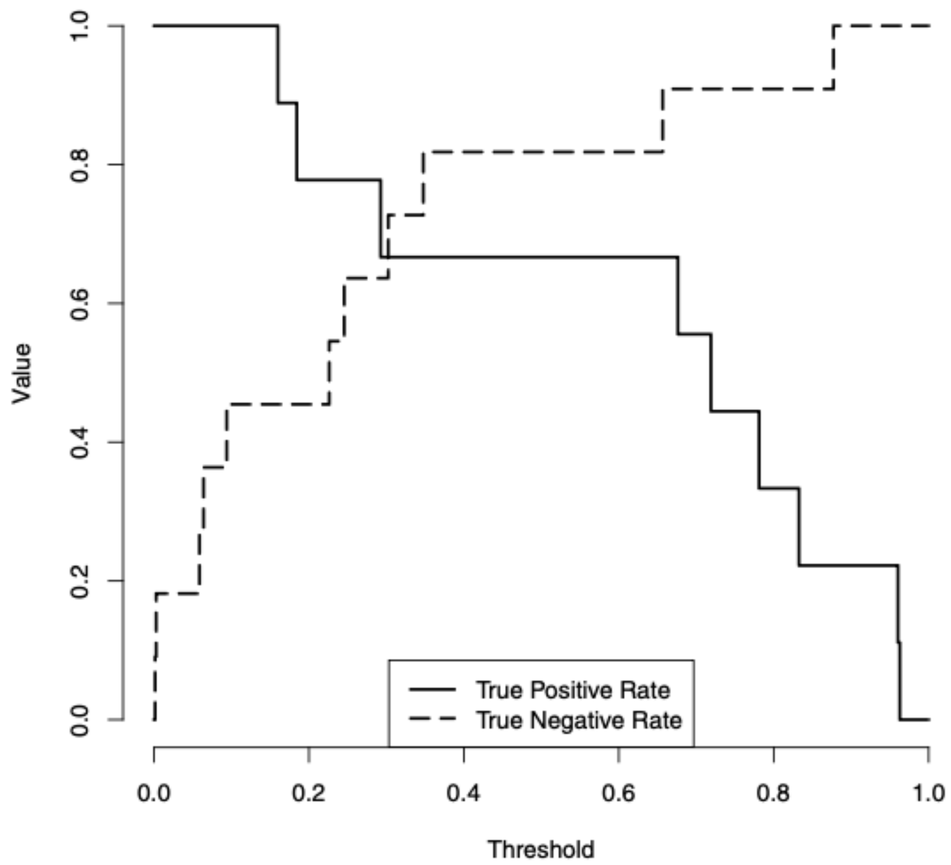
Value in a range

Hence, we use a threshold to map the score into an estimated target value.

$$\text{threshold}(\text{score}, 0.5) = \begin{cases} 1 & \text{if score } \geq 0.5 \\ 0, & \text{otherwise} \end{cases}$$

How to Choose the Threshold on Prediction Scores?

The threshold can be controlled, hence control accordingly, TPR and TNR.

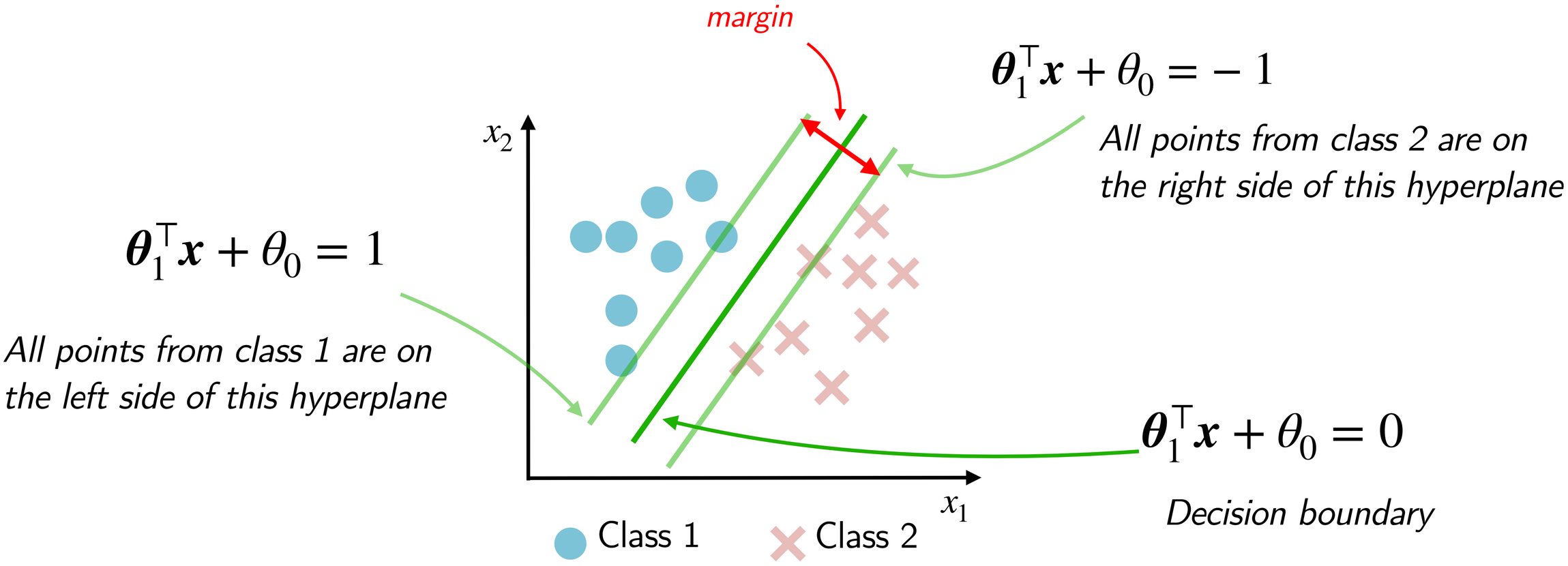


For every value of threshold, we have different values of TPR and TNR.

There a trade-off between accuracy for predictions or positive target levels and accuracy for prediction of negative target levels.

figure from ML for Predictive Data Analytics textbook.

SVM Separability Condition Visualization (Separable Case)

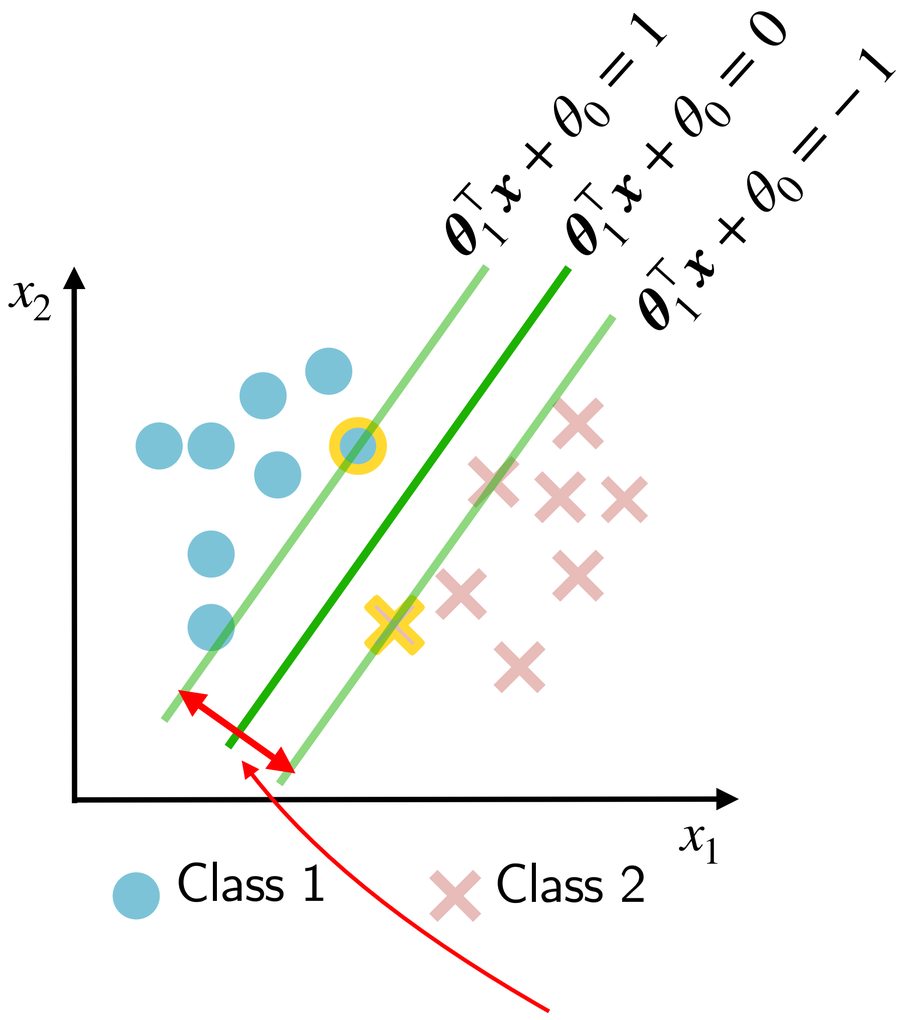


SVM Margin Visualization

To find a classifier that maximize the margin,

$$\exists\ \boldsymbol{x}_i^+ \text{ s.t. } c(\boldsymbol{x}_i^+) = +1 \quad \text{and} \quad \boldsymbol{\theta}_1^\top \boldsymbol{x}_i^+ + \theta_0 = +1$$

$$\exists\ \boldsymbol{x}_i^- \text{ s.t. } c(\boldsymbol{x}_i^-) = -1 \quad \text{and} \quad \boldsymbol{\theta}_1^\top \boldsymbol{x}_i^- + \theta_0 = -1$$



Intuition: The support hyperplane$^+$ touch one example from c_1 and hyperplane$^-$ touch one example from c_2.

How to quantify the margin?

Why? Otherwise, there will still be room to increase (maximize) the margin (distance between the two hyperplanes $\{\boldsymbol{\theta}_1^\top \boldsymbol{x} + \theta_0 = 1\}$ and $\{\boldsymbol{\theta}_1^\top \boldsymbol{x} + \theta_0 = -1\}$).

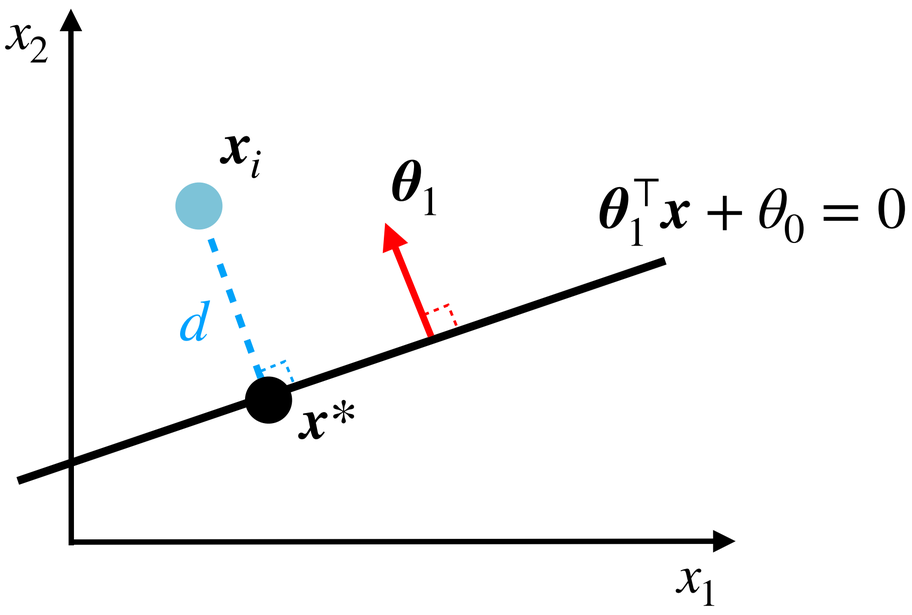
Distance from a Point to a Hyperplane (I)

Intuition: $\boldsymbol{x}_i - \boldsymbol{x}^*$ is along the direction of $\boldsymbol{\theta}_1$.

Given the hyperplane $\mathrm{H} = \{\boldsymbol{\theta}_1^\top \boldsymbol{x} + \theta_0 = 0\}$,

the point, on the hyperplane, closest to $\boldsymbol{x}_i$ is

$$\boldsymbol{x}^* = \arg\min_{\boldsymbol{x} \in \mathrm{H}} \|\boldsymbol{x}_i - \boldsymbol{x}\|_2$$



If $\boldsymbol{x}_A, \boldsymbol{x}_B \in \mathrm{H}$, then $\langle \boldsymbol{x}_i - \boldsymbol{x}^*, \boldsymbol{x}_A - \boldsymbol{x}_B \rangle = 0$, i.e.,

$$\exists\ \beta \text{ s.t. } \boldsymbol{x}_i - \boldsymbol{x}^* = \beta \boldsymbol{\theta}_1$$

$$\boldsymbol{x}_i - \boldsymbol{x}^* = \beta \boldsymbol{\theta}_1 \quad \Rightarrow \quad \boxed{d := \|\boldsymbol{x}_i - \boldsymbol{x}^*\| = \|\beta \boldsymbol{\theta}_1\|}$$

Continue next slide to derive β and margin.

SVM Optimization for Non-Separable Case

Non-separable case:

$$\min_{\boldsymbol{\theta}} \ \frac{1}{2}\|\boldsymbol{\theta}_1\|_2^2 \ + \ C\sum_{i=1}^{n}\zeta_i$$

$$\text{s.t.} \quad y_i(\boldsymbol{\theta}_1^\top \boldsymbol{x}_i + \theta_0) \geq 1 - \zeta_i, \quad \text{for } i = 1,\ldots,n$$

$$\zeta_i \geq 0, \quad \text{for } i = 1,\ldots,n$$

We define a cost variable $\zeta > 0$ to penalize points inside the margin or on the wrong side.

Depending on the cost C to pay attention to the examples with highest C.